

Effective treatment of systemic candidiasis by synergistic targeting of cell wall synthesis

Received: 11 January 2025

Accepted: 2 June 2025

Published online: 01 July 2025



Ju Yeon Chung^{1,5}, Yoon-Kyoung Hong^{2,5}, Eunhee Jeon^{1,2}, Seungju Yang¹,
Ayoung Park¹, Ralph Weissleder^{3,4}, Yong Pil Chong^{1,2}✉ &
Hyun Jung Chung¹✉

Fungal infections pose a serious threat to global human health fueled by the increase in immunosuppressive therapies, medical implants, and transplantation. The emergence of multidrug resistance with limited options of current antifungal drugs are a further constraint. There is thus a clear and unmet need to identify therapeutic targets and develop alternative classes of antifungal agents. Here, we hypothesize that dual targeting of key regulatory genes of fungal cell wall synthesis (*FKS1* encoding β -1,3-glucan synthase and *CHS3* encoding chitin synthase) can synergistically inhibit fungal growth. Based on iterative designs, we generate a small library of fungal-targeted nanoconstructs, and identify a lead construct (FTNx) that shows preferential accumulation in fungal cells over mammalian cells and leads to prominent antifungal effects in vitro. We further show that FTXn is highly effective in a mouse model of disseminated candidiasis, demonstrating diminished fungal growth and enhanced survival rate. This strategy appears promising as an effective treatment for fungal infections in mammalian hosts.

Fungal infections caused by *Candida* pathogens pose a significant health risk, accounting for some of the most severe and challenging hospital-acquired infections^{1–3}. Despite the high mortality and morbidity rates associated with fungal infections in humans, the current treatment options are inadequate, and the discovery pipeline for antifungal drugs is underdeveloped⁴. The fungal cell wall, consisting of unique and essential components of fungal viability, presents a promising drug target⁵. For instance, the cell wall of *Candida albicans* consists of chitin (2%), β -1,3-glucan (62%), and mannan (36%), along with other distinctive structural components⁶. Currently available antifungal drugs primarily target the fungal cell wall, including azoles that inhibit ergosterol biosynthesis, polyenes that directly bind to ergosterol, and echinocandins that block fungal β -1,3-glucan biosynthesis^{7,8}. Systemic candidiasis, an invasive and difficult-to-treat form of candidiasis, poses significant challenges due to the poor fungal

targeting ability and the pronounced off-target tissue toxicity of existing drugs that limits their effective doses^{9,10}. Given the current limitations in treatment options, the emergence of multidrug resistance, and the high toxicity of existing antifungal agents^{11–13}, there is a pressing need to identify therapeutic targets and develop alternative classes of antifungal agents^{14–16}.

Antisense oligonucleotides (ASOs) have shown great promise as therapeutics, acting through targeted mRNA degradation or splicing modulation of specific genes¹⁷. For antifungal therapy, targeting of the biosynthesis of essential fungal cell wall components can provide an alternative treatment option to conventional antifungal agents^{6,18}. To achieve efficacy in vivo, strategies for the selective delivery of ASOs to target cells are crucial¹⁹. While lipid nanoparticles (LNPs) have proven effective for liver diseases and vaccines^{20–22}, they exhibit poor targeting ability for other targets, including microorganisms such as fungal cells²³. Since fungal cells consist of a thick cell wall with distinct

¹Department of Biological Sciences, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea. ²Department of Infectious Diseases, Asan Medical Center, University of Ulsan College of Medicine, Seoul, Republic of Korea. ³Center for Systems Biology, Massachusetts General Brigham, Boston, Massachusetts, USA. ⁴Department of Systems Biology, Harvard Medical School, Boston, Massachusetts, USA. ⁵These authors contributed equally: Ju Yeon Chung, Yoon-Kyoung Hong. ✉e-mail: drchong@amc.seoul.kr; hyunjc@kaist.ac.kr

components, the delivery of nucleic acids to fungal pathogens has been rarely attempted and limited to *in vitro* studies^{24–26}.

Here, we hypothesize that dual targeting of the biosynthesis of essential fungal cell wall components, β -1,3-glucan and chitin, can potentially synergize and result in the effective treatment of candidiasis. The two essential cell wall components are synthesized and regulated by key enzymes encoded by *FKS1* for β -1,3-glucan and *CHS3* for chitin^{18,27,28}. Given the absence of dual-targeting small molecules with acceptable safety profiles, we use different fungal-targeted antisense oligonucleotides (*fso*) to target the cell wall synthesis genes. For fungal cell targeting, we construct a small, targeted library of nanoconstructs for *fso* delivery. The nanoconstructs consist of a polymer-stabilized gold core, charged polymers that can condense the nucleic acids while interacting with the fungal cell wall (*X*), and *fso* for dual gene targeting of *FKS1* and *CHS3* (Fig. 1). We initially perform comparative screens to identify the formulation FTNx (fungal-targeted nanoconstructs), which exhibit preferential uptake into fungal cells over mammalian cells, likely due to carbohydrate polymer-polymer interactions. We then successfully identify several lead FTNx constructs that induce synergistic antifungal effects, which are subsequently tested for efficacy upon *in vivo* delivery. Using mouse models of disseminated candidiasis, we demonstrate marked antifungal efficacy and prolonged survival of mice upon treatment with FTNx. The current strategy thus presents a potent and alternative treatment option that effectively targets human fungal pathogens and addresses the challenges of antifungal resistance.

Results and discussion

Generation of nanoconstruct library for fungal delivery

To establish the FTNx platform, we constructed a small library of cationic gold nanoconstructs (5 nm core) differing in composition, leading to various particle size, charge, *fso* loading capacity, and polarity (Supplementary Table 1). The nanoconstruct consisted of cationic polymer-modified gold nanoparticle cores (NPs)^{29–32}, a secondary polymer cation (*X*), and *fso*, to produce Nx(*X*). The secondary cation was necessary to endow a positive charge to the formulation,

which was generally required to enable uptake into the cells^{33,34}. Various secondary polymeric cations such as chitosan (CS), polyethyleneimine (PEI), poly(allylamine) (PAA), poly(amidoamine) (PAM), as well as monomeric cations spermine (SPN), and protamine (PTN) were explored and tested (Fig. 2a). Dynamic light scattering measurements (DLS) revealed that the hydrodynamic diameters of the nanoconstructs ranged from approximately 48 nm to 158 nm, which varied according to the cation (*X*) (Fig. 2b). Among these, CS resulted in particles with sizes larger than 130 nm, while PEI, PAA, and PTN gave rise to smaller particles (< 100 nm). Sizes varied for PAM with different molecular weights, while SPN resulted in medium-sized particles. CS with low (5 kDa, CS_{low}) and high (190–310 kDa, CS_{high}) molecular weights exhibited the lowest polydispersity indices (PDI), that were 0.19 and 0.17, respectively. According to zeta potential measurements, nanoconstructs prepared with branched and linear PEI (BPEI and LPEI), CS, PAA, and SPN exhibited positive charges that ranged from +19.3 mV to +68.4 mV, while ones prepared with PAM were negatively charged ranging from −31.9 mV to −43.5 mV (Fig. 2c). The negative zeta potential observed for the particles prepared using *fso* can either be due to the limitation of PAM interacting with the *fso*-bound GNPs, or with the free *fso* after displacement, determining that the polymer could not efficiently load the *fso* and form stable nanoconstructs.

To explore fungal targeting, we performed microscopy experiments in which different Nx(*X*) preparations, including fluorescein amidite (FAM)-labeled *fso* were incubated with fungal (*C. albicans*) and mammalian (NIH-3T3) cells. In fungi, we observed two distinct staining patterns: (i) preferential cell wall staining (BPEI, LPEI, PAA, PAM, SPN, and PTN) or (ii) punctuate patterns in fungal cytoplasm (CS_{low}, CS_{med}, CS_{high}). Since we were primarily interested in *fso* delivery to the inside of fungal cells, we focused on formulations that exhibited the punctuate intracellular staining pattern. Of these, formulations prepared with CS_{low} showed the highest fungal uptake (Fig. 2d, f)^{35,36}. The foci patterns observed specifically for CS_{low} indicate that internalization occurred by endocytosis, resulting in the formation of endosomes in the fungal cells. Examining *in vitro* release of *fso* from the nanoconstructs formed with CS_{low} showed that 83% cumulative release was

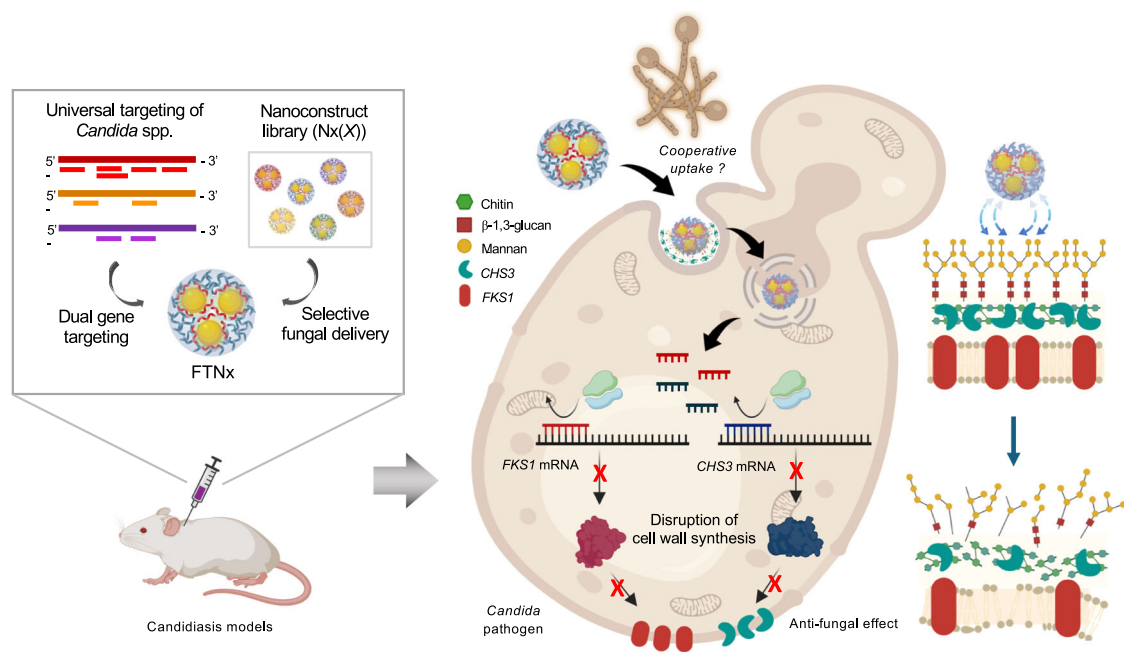


Fig. 1 | Overview of antifungal therapy design and experimental approach. We generated a small nanoconstruct library (Nx(*X*)) to condense fungal silencing oligonucleotides (*fso*) for synergistic targeting of fungal cell wall synthesis. The design involved the formation of nanoconstructs that selectively interact with the fungal

cell wall components and induce internalization, leading to FTNx. Dual gene silencing (*FKS1* and *CHS3*) by FTNx leads to impaired synthesis of β -1,3-glucan and chitin, resulting in prominent antifungal effects *in vivo*. Created in BioRender. Chung, J. (2025) <https://BioRender.com/9q3cdpr>.

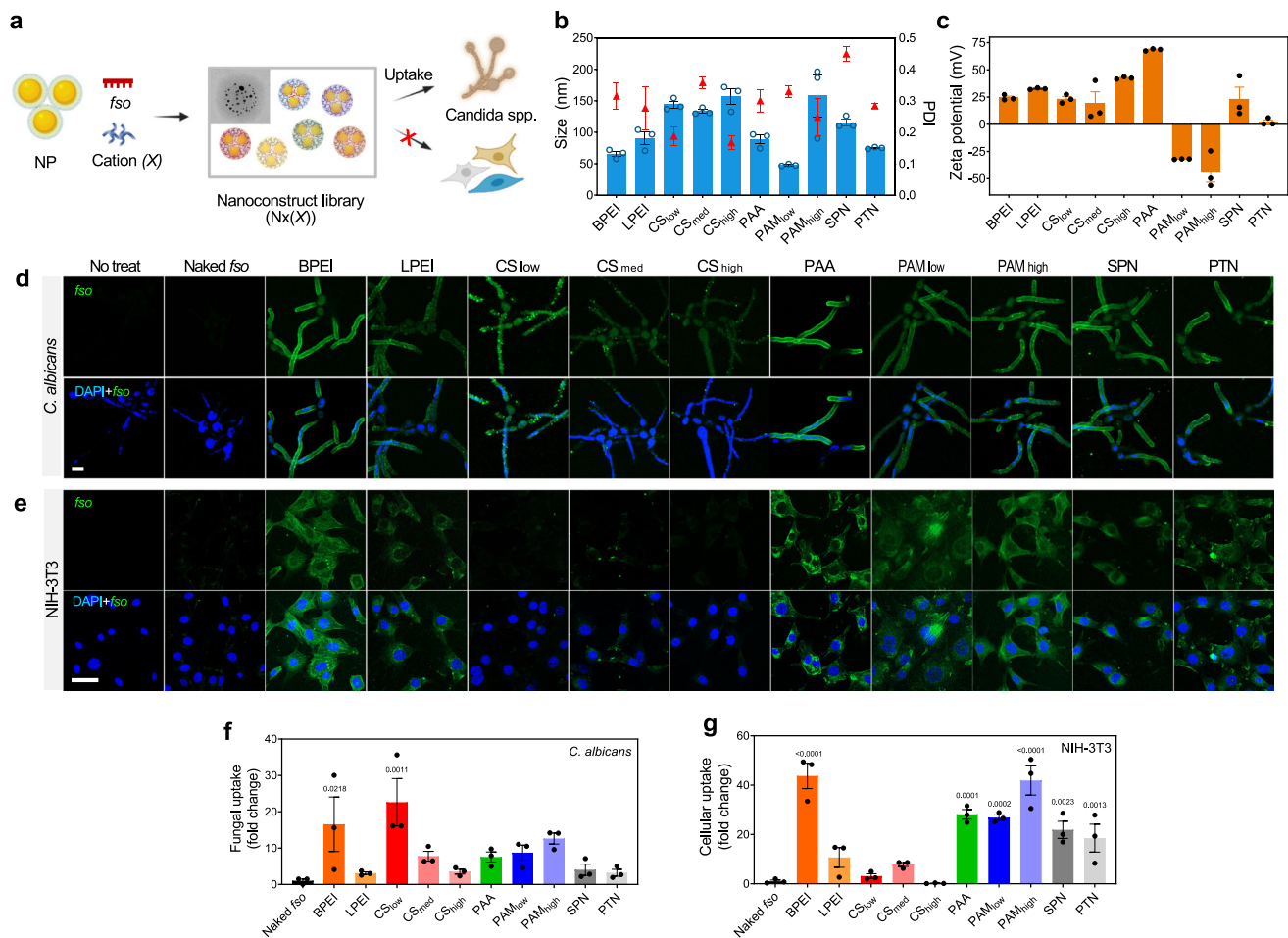


Fig. 2 | Development of *fso*-delivering nanoconstructs for fungal targeting.

a Preparation of *fso*-loaded nanoconstructs (Nx(X)) uses various cations (X) for complexation to stabilize the gold nanoparticle cores, that were screened for formulations that have preferential fungal uptake and minimal mammalian cell uptake. **b** DLS and **c** zeta potential measurements of Nx(X). Data represent mean $\pm$ s.e.m. ($n = 3$ biological replicates). **d, e** Representative confocal images of *C. albicans* (**d**, scale bar, 5 μ m) and NIH-3T3 cells (**e**, scale bar, 50 μ m) treated with

Nx(X) or naked *fso* as the control ($n = 3$ biological replicates; green, FAM-labeled *fso*; blue, 4',6-diamidino-2-phenylindole (DAPI)). **f, g** Quantification of uptake in *C. albicans* (**f**) and NIH-3T3 cells (**g**) obtained from (**d**) and (**e**). Data represent mean $\pm$ s.e.m. ($n = 3$ biological replicates). One-way ANOVA used to obtain the p -value by comparison of groups with the naked *fso* group. Representative data from three independent experiments. Source data are provided as a Source Data file. **a** created in BioRender. Chung, J. (2025) <https://BioRender.com/9q3cdpr>.

achieved at 3 h (Supplementary Fig. 1). Overall, these results indicate that nanoconstructs prepared with CS_{low} were able to internalize into the fungal cells, while the others could not and were primarily bound to the cell membrane. We attribute the unique fungal internalization of the formulation with CS_{low} to charge interactions with the abundant fungal β -1,3-glucan, which can also compete with the fungal chitin components^{37–39}. When we performed the uptake experiments with fibroblasts, strong *fso* signals appeared in the cytoplasm with most formulations except for the ones with CS (Fig. 2e, g).

Optimization and fungal delivery of FTNx

We further examined the FTNx platform by additional optimization and characterization experiments (Supplementary Fig. 2a). DLS measurements showed that the size of NP was 54 nm, which increased to 106 nm upon the addition of *fso*, and to 143 nm after addition of CS_{low} (Supplementary Fig. 2b). The zeta potential increased from +4.1 mV to -9.4 mV upon adding NP with *fso*, while the subsequent addition of CS_{low} increased the value to +37.0 mV (Supplementary Fig. 2c). Varying the CS content showed that medium CS content (N/P ratio = 0.1) resulted in the most stable particles (Supplementary Fig. 3). Observation through transmission electron microscopy (TEM) showed that the finally prepared FTNx was -150 nm in mean diameter and appeared as uniform particles including clusters of the core NPs, which can form by

electrostatic interactions of the cationic NP surface, *fso*, and CS (Supplementary Fig. 4)^{40,41}. Notably, the exclusion of either CS or *fso* from FTNx resulted in irregular particles with undefined shapes. Scanning electron microscopy (SEM) also confirmed the observations, showing well-defined shapes and uniform size distributions of FTNx (Supplementary Fig. 5). Long-term size measurements showed that the FTNx formulation remained stable for up to 7 days (Supplementary Fig. 6).

We next validated the delivery potential of FTNx compared with the conventional formulations. Upon treatment to *C. albicans*, FTNx resulted in the strongest *fso* signals within the fungal cytoplasm, while the other formulations resulted in signals that were either too low or confined to the surface and outside the cells (Supplementary Fig. 7a). Particularly, fungal cells treated with complexes of free PEI (PEI/*fso*) or lipid nanoparticles (LNPs) showed the presence of aggregates attached to the surface of the fungal cell wall that could not enter the cytoplasm. Complexes formed with free CS (CS/*fso*) also remained localized to the cell wall, while the cytoplasm was void of signals. Formulations using silver nanoparticles (silver NC) or liposomes did not show significant uptake into the fungal cells. On the other hand, FTNx showed neglectable uptake into NIH-3T3 cells, while LNP, liposomes, PEI/*fso*, and CS/*fso* showed substantial uptake (Supplementary Fig. 7b). Quantification of fluorescence determined that FTNx exhibited the highest uptake in *C. albicans* and minimal uptake in NIH-3T3 cells,

while LNP showed only modest uptake in *C. albicans* but high uptake in NIH-3T3 cells (Supplementary Fig. 7c, d). These results reveal that only FTNx could be effectively delivered into fungal cells, that can be due to the physicochemical properties of NPs as well as the function of CS that facilitates fungal cell entry^{38,39,42}. Particularly, the preferential uptake of FTNx can be majorly attributed to the competition between the CS of FTNx and fungal chitin on binding with β -1,3-glucan. Among CS with various molecular weights, CS_{low} (Mw 5000) would have the optimal physicochemical properties leading to permeation through the fungal cell wall^{36,37}. The amount of CS also affected fungal uptake, with FTNx prepared at 2 μ g/ml CS resulting in the highest internalization into *C. albicans* and minimal uptake in NIH-3T3 cells (Supplementary Fig. 8). Increasing the CS concentration beyond this level led to a decrease in fungal uptake, likely due to the competition of unbound CS with the nanoconstructs, or an increase in particle size, which should impair the uptake ability. Factors that affect fungal and mammalian cell uptake include surface charge, particle size, and the biochemical interactions of the material with the cell wall or membrane. Previous studies have reported that particles with strong cationic and lipophilic properties result in higher uptake into mammalian cells^{43,44}. Here, delivery into the fungal cytoplasm appeared to involve specific biochemical interactions of the material with the cell wall components. The above results led to the optimized formulation of FTNx that would enable effective and selective delivery of *fso* to *Candida* pathogens, while avoiding off-target internalization into host cells upon in vivo delivery.

Antifungal effect by gene targeting with FTNx in vitro

We next asked whether antifungal effects of FTNx could occur (i) via silencing cell wall synthesis genes by functional *fso* delivery, and (ii) by dual *fso* delivery leading to impaired cell wall synthesis resulting in fungal cell death. To explore these questions, we selected target gene sequences of *FKS1* and *CHS3* of the standard strain *C. albicans* (Ca) (Fig. 3a). *FKS1* encodes the enzyme for synthesis of β -glucan, which serves as the primary scaffold of the cell wall, while *CHS3* encodes the enzyme for chitin synthesis, which binds with β -glucan that is further bound to mannan and reinforces the cell wall structure^{27,28}. *C. albicans* has a compensatory mechanism where chitin synthesis is upregulated when glucan synthesis is inhibited, therefore dual inhibition of the two targets can provide synergistic antifungal effects. To avoid any possible interference in the gene silencing function, all experiments assessing the efficacy of *fso* were conducted without modification with FAM. Treatment of *C. albicans* with FTNx delivering *fso* targeting either *FKS1* (F.Ca) or *CHS3* (C.Ca), resulted in a decrease in fungal growth, while targeting *FKS1* was more potent compared to *CHS3* (Supplementary Fig. 9a–c)^{45–47}. Measuring target mRNA expression levels showed that the treatment with FTNx resulted in significant reductions in the expression of *FKS1* and *CHS3*, demonstrating that mRNA degradation by *fso* majorly contributed to the antifungal effect (Fig. 3b, c)⁴⁸. We compared the *fsos* targeting fungal genes with non-target *fso* in the FTNx platform, shown as FTNx (NT). The non-target *fso* was designed based on a comparison of sequences of *C. albicans*, *C. auris*, *C. glabrata*, *C. parapsilosis*, and *C. tropicalis*, and selection of ones with 0% homology.

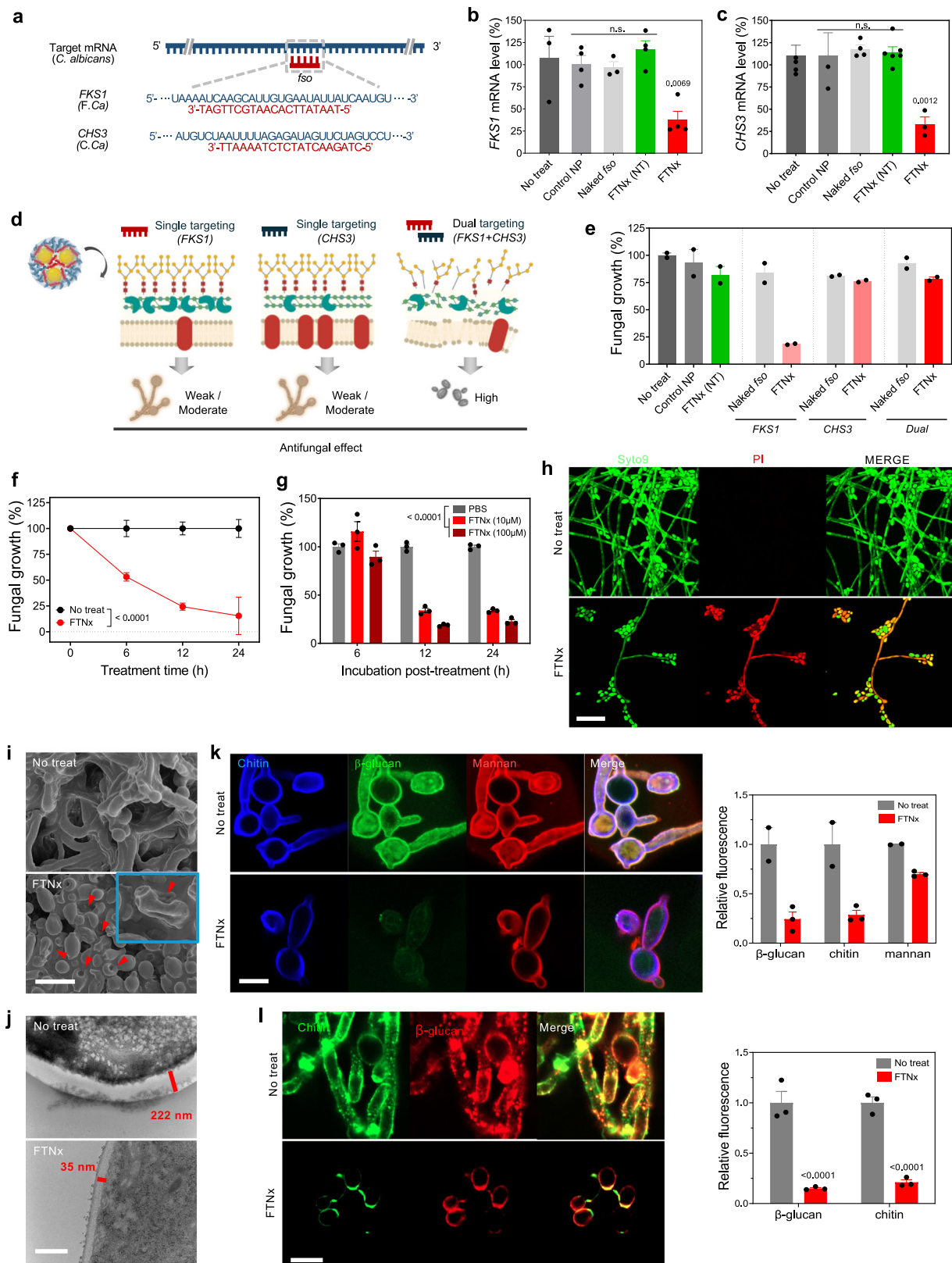
We next investigated whether dual *fso* delivery targeting *FKS1* and *CHS3* by FTNx could exert a synergistic effect in antifungal activity (Fig. 3d). We reasoned that simple *FKS1* inhibition could lead to impaired cell wall synthesis but also trigger post-transcriptional regulation of *CHS3* to restore chitin deposition, leading to only weak to moderate antifungal effects^{47,49,50}. Thus, silencing both *FKS1* and *CHS3* could block the restoration, leading to cell wall disruption and effective fungal cell death. We precisely observed this effect and show that dual *fso* targeting *FKS1* (F.Ca) and *CHS3* (C.Ca) lead to efficient inhibition of fungal growth (Fig. 3e). Varying the incubation time of FTNx delivering dual *fso* resulted in increased antifungal effects with longer

incubation (Fig. 3f). Increasing the concentration of *fso* loading resulted in superior antifungal effects (Fig. 3g). Visualization of the live and dead fungal cells confirmed that the majority of the fungal cells treated with FTNx were dead, while ones without treatment were mostly alive, demonstrating the fungicidal effect of FTNx (Fig. 3h and Supplementary Fig. 10). Furthermore, the morphology of the fungal cells treated with FTNx exhibited shortened or absence of pseudohyphae, whereas the controls showed the presence of long, extended pseudohyphae, which is indicative of fungal virulence⁵¹. These morphological changes could also be observed by SEM (Fig. 3i and Supplementary Fig. 11a). High-resolution TEM analyses revealed that dual *fso* delivery by FTNx resulted in a spore-like phenotype, with structural defects and reduced thickness of the cell wall compared to the control (Fig. 3j and Supplementary Fig. 11b). Immunocytochemistry also demonstrated that *C. albicans* treated with dual *fso*-loaded FTNx exhibited significantly reduced deposition of β -1,3-glucan and chitin, compared to the control (Fig. 3k and Supplementary Fig. 12). Quantification of the fluorescence signals demonstrated that β -glucan, chitin, and mannan levels were reduced to 24, 28, and 70% in *C. albicans* treated with FTNx, compared to the untreated controls. Treating FTNx to *S. cerevisiae* as negative control yeast cells did not change β -glucan, chitin, or mannan levels compared to the untreated controls (Supplementary Fig. 13). Wheat germ agglutinin (WGA) staining also showed a significant reduction in chitin signal for FTNx treatment compared to the control, whereas *S. cerevisiae* showed no significant changes upon treatment (Fig. 3l and Supplementary Fig. 14). Therefore, dual gene silencing by FTNx exerted a synergistic effect in antifungal activity, by inhibiting the synthesis of multiple cell wall components simultaneously. We measured the minimum inhibitory concentration (MIC) of *C. albicans* which was 2.5 mg/L, and the lethal dose (LD50) of NIH-3T3 was 23783.4 mg/L, resulting in a selectivity index of 9513.36 for mammalian cells over fungal cells (Supplementary Fig. 15). These results demonstrate the high selectivity of FTNx, providing an alternative treatment option to conventional antifungal agents.

Universal targeting of multiple *Candida* species

C. albicans is the most common fungal species in humans, but infections with *C. tropicalis*, *C. parapsilosis*, *C. glabrata*, *C. krusei*, and *C. auris* are also significantly observed. We therefore were interested in designing candidate *fso* sequences that target conserved regions of *FKS1* or *FKS2*, and *CHS3* genes in all of these species (Fig. 4a). Among the candidates, F.57–58 for *FKS1* and C.82–83 for *FKS2* showed the most potent antifungal effects (Fig. 4b). We then evaluated the antifungal effect of FTNx delivering different combinations of dual *fso*. While some of the universally designed *fso* exhibited lower antifungal effects compared to the *C. albicans*-specific *fso*, F.48 and F.57–58 for *FKS1* and C.57 for *CHS3* showed superior antifungal effects (Fig. 4c). Time-lapse measurements showed that dual *fso* delivery of selected universally designed candidates F.48 + C.57 resulted in an antifungal effect that was evident at as early as 6 h and lasted until 24 h post-treatment (Fig. 4d). Increasing the concentration of FTNx resulted in stronger antifungal effects (Fig. 4e). We next tested *fso* F.48 + C.57 in *C. tropicalis*, *C. glabrata*, and *C. krusei*, which showed efficient uptake by FTNx, while naked *fso* did not (Fig. 4f). Treatment of seven different strains of five *Candida* species with FTNx delivering the universally designed *fso* showed significant antifungal effects in all seven of the strains (Fig. 4g). These results demonstrate that the FTNx platform can be versatily applied to multiple *Candida* pathogens using a universal *fso* for each target gene. FTNx can also be potentially applied to various fungi other than *Candida* that cause human infections. Examples are yeast-like fungi, such as *Cryptococcus neoformans* and *Malassezia* spp, and filamentous fungi, such as *Aspergillus* spp., which exhibit similar cell wall structures to *Candida*^{52–54}.

To maximize the efficacy for further validation in vivo, we investigated the use of the gapmer platform, which consisted of a



phosphorothioate (PS) DNA backbone flanked with 2'-O-methyl RNA at both ends^{55,56}. Treating the gapmers of the *fso* candidates for *FKS1* in the presence of micafungin to *C. albicans* in vitro, showed that the majority of the gapmers were superior to the unmodified *fso*, with candidates F.54 and F.57 being the most potent in antifungal activity (Fig. 4h). For

CHS3, the gapmers of *C.Ca.* and C.57 showed slight enhancement compared to the unmodified form. Dual *fso* treatment using the combination of gapmers of either F.54 or F.57 for *FKS1* along with *C.Ca.* for *CHS3* resulted in major inhibition of fungal growth in vitro, with F.57 + *C.Ca.* showing the strongest effect (Supplementary Fig. 16).

Fig. 3 | Dual gene targeting of cell wall synthesis is an effective antifungal strategy. **a** Sequences of *fso* targeting *FKS1* and *CHS3* specific to *C. albicans*. **b, c** mRNA expression levels of *FKS1* (**b**, $n = 3$, no treat and naked *fso*; $n = 4$, control NP, FTNx(NT), and FTNx), and *CHS3* (**c**, $n = 5$, no treat; $n = 3$, control NP and FTNx; $n = 4$, naked *fso*; $n = 6$, FTNx(NT)) in *C. albicans* treated with FTNx (100 nM), control NP (100 nM), naked *fso* (50 μ M), or FTNx(NT) (100 nM), normalized to levels of *ACT1* mRNA. Data represent mean $\pm$ s.e.m. from biological replicates. One-way ANOVA for comparison with 'No treat' control. **d** Schematic of the mechanism of antifungal activity by dual gene silencing of *FKS1* and *CHS3* in *Candida* spp. Targeting a single gene leads to weak to moderate effects, while dual gene targeting leads to potent antifungal effects. **e–g** Antifungal effect of FTNx in *C. albicans* measured by the WST-8 assay. **e** Comparison of single or dual *fso* with FTNx (100 nM, 24 h, $n = 2$). **f** Various time periods (100 nM, $n = 3$), and **g** different doses

with incubation times ($n = 3$) on treating FTNx with dual *fso*. **h** Confocal microscopy of live and dead fungal cells stained with SYTO9 and PI, respectively (scale bar, 20 μ m). **i** SEM (scale bar 10 μ m), and **j** TEM (scale bar 400 nm) of *C. albicans* treated with FTNx with dual *fso*. **k, l** Fluorescence staining of cell wall components in *C. albicans* treated with FTNx and quantification by normalization to 'No treat' control. **k** Blue, chitin (CaFW); green, β -1,3-glucan (Dectin-488); red, mannan (ConA-Tred); scale bar, 5 μ m; $n = 2$ for no treat, $n = 3$ for FTNx. **l** Green, chitin (WGA-488); red, β -1,3-glucan (Dectin-647); scale bar, 5 μ m; $n = 3$. **e–l** Data represent mean $\pm$ s.e.m. from biological replicates (Two-way ANOVA for comparison with No treat control). Representative data from three independent experiments. Source data are provided as a Source Data file. **a, d** created in BioRender. Chung, J. (2025) <https://BioRender.com/9q3cdpr>.

In vivo efficacy of FTNx in mouse models of *Candida* infection

To further estimate the pharmacokinetics and dynamics of the FTNx construct, we performed biodistribution and therapeutic efficacy studies in mouse models^{57,58}. We first assessed the biodistribution of FTNx in uninfected mice by fluorescence imaging (Fig. 5a). At 6 h post-injection, the FTNx group exhibited strong *fso* signals, especially in the kidney, liver, and lung tissues, that were much higher compared to naked *fso* treatment (Fig. 5b, c). This was due to the delayed clearance of *fso* in the nanoformulation, leading to extended half-life, compared to naked *fso*^{59,60}. When examining biodistribution of FTNx in an infected mouse model with *C. albicans*, similar patterns were observed with detectable signals in the kidney, liver, and lungs (Supplementary Fig. 17).

We next evaluated the therapeutic efficacy of FTNx in a systemic candidiasis model with cyclophosphamide-induced immunosuppression, similar as often seen in the clinic (Fig. 5d). To determine the optimal dose, we measured CFU levels in the kidney at various doses. Our data show that IX dose treatment of FTNx delivering dual *fso* (F.Ca + C.Ca., Fig. 5e) completely abrogated fungal growth, whereas FTNx including non-target *fso* (NT) did not induce any noticeable antifungal effects (Fig. 5f). We then examined FTNx delivering gapmers of dual *fso*, which included a universally designed *fso*, as a potential multi-target therapeutic of *Candida*. Results showed that FTNx delivering *fso* (F.57 + C.Ca) was able to eradicate the infection, while FTNx including the NT control could not (Fig. 5g). Histological analyses of kidney tissues by periodic acid-Schiff (PAS) and hematoxylin and eosin (H&E) also confirmed the major antifungal effect by FTNx (F.57 + C.Ca) (Fig. 5h). We also tested the efficacy of FTNx treated at a later time point, that is, 6 h post-infection, followed by additional treatment after 6 h. Remarkably, no detectable fungal burden was observed in the FTNx-treated group, in stark contrast to the PBS control group, which exhibited a high level of fungal colonization (Supplementary Fig. 18). This result underscores the therapeutic potential of the FTNx platform even under delayed treatment conditions. Most impressive were the survival benefits, showing 100% survival in the FTNx group with the lethal candidiasis model (Fig. 5i). Histological analysis at day 7 showed that FTNx significantly reduced fungal growth compared to the PBS control, although some extent of fungal growth appeared due to constraint in dose of the treatment in the acute infection model (Supplementary Fig. 19). The average body weights of mice treated with FTNx did not significantly change (Fig. 5j).

We also investigated FTNx in a second mouse model of dermal candidiasis (Fig. 5k)⁵⁸. We again saw diminished fungal growth and minimal inflammation using the lead FTNx (Fig. 5l, m). Conversely, the PBS control group showed severe fungal growth with pseudo-hyphae formation and signs of inflammation. To evaluate in vivo toxicity of FTNx, we measured the serum levels of alanine aminotransferase (ALT) activity and blood urea nitrogen (BUN), which showed that the levels were within the normal range (28–132 U/L for ALT, and 18–29 mg/dL for BUN), determining minimal in vivo toxicity (Fig. 5n, o)⁶¹.

In sum, these results provide evidence that dual gene silencing of fungal cell wall synthesis using the FTNx platform can completely diminish fungal growth in vivo and alleviate candidiasis, while exhibiting negligible toxicity. We would like to note that challenges can occur related to the delivery efficiency of FTNx to necrotic areas, which are associated with poor vascularization, hypoxia, and acidic conditions. Strategies to overcome these barriers should be explored in the future. Particularly, *Candida* pathogens such as *C. auris* has been reported to exhibit high resistance to conventional antifungals, suggesting that the current platform can provide an alternative treatment option. Further engineering of the nanoconstruct platform can endow targeted delivery and efficacy against other pathogens, such as difficult-to-treat molds, which thrive in hypoxic environments and resist standard antifungals.

In conclusion, we present a therapeutic platform for treating *Candida* infections using a fungal-targeted dual gene silencing approach that disrupts cell wall synthesis of the pathogens. The FTNx platform facilitated the effective and selective delivery of *fso* targeting genes *FKS1* and *CHS3* to fungal pathogens, resulting in significant antifungal effects. By employing dual *fso* delivery with FTNx, we were able to completely eliminate systemic candidiasis and inhibit fungal growth in skin infection in mice. This approach holds great promise as an effective therapeutic platform for managing fungal infections caused by *Candida* spp., offering an alternative treatment option to conventional antifungals. Conventional antifungal agents such as micafungin are highly potent and widely used in the clinic, however their high toxicity due to off-target effects and the emergence of resistance pose major risks^{62,63}. The FTNx platform, enabling multiple targeting in a single formulation while causing minimal off-target effects, presents a promising alternative. Although widespread application of GNPs in the clinic has been somewhat limited, further investigation on large-scale preparation as well as fully understanding their biodistribution and clearance patterns will provide opportunities for translation. We anticipate that the FTNx platform can be applied to various fungal pathogens by targeting various gene targets, which could potentially overcome resistance problems associated with these infections.

Methods

Ethics

All experimental protocols were approved in accordance with the policies of the Institutional Animal Care and Use Committee of Asan Medical Center (#2022-12-260) and KAIST (#KA2023-076-v1). All research involving animal subjects was performed in accordance with relevant ethical guidelines.

Materials

Gold nanoparticles functionalized with N-hydroxysuccinimide (NHS) ester (5 nm, ~1 NHS group per nm²) were acquired from Cytodiagnostics (CGNK-5-2). BPEI (Mw 25 kDa, 408727) and LPEI

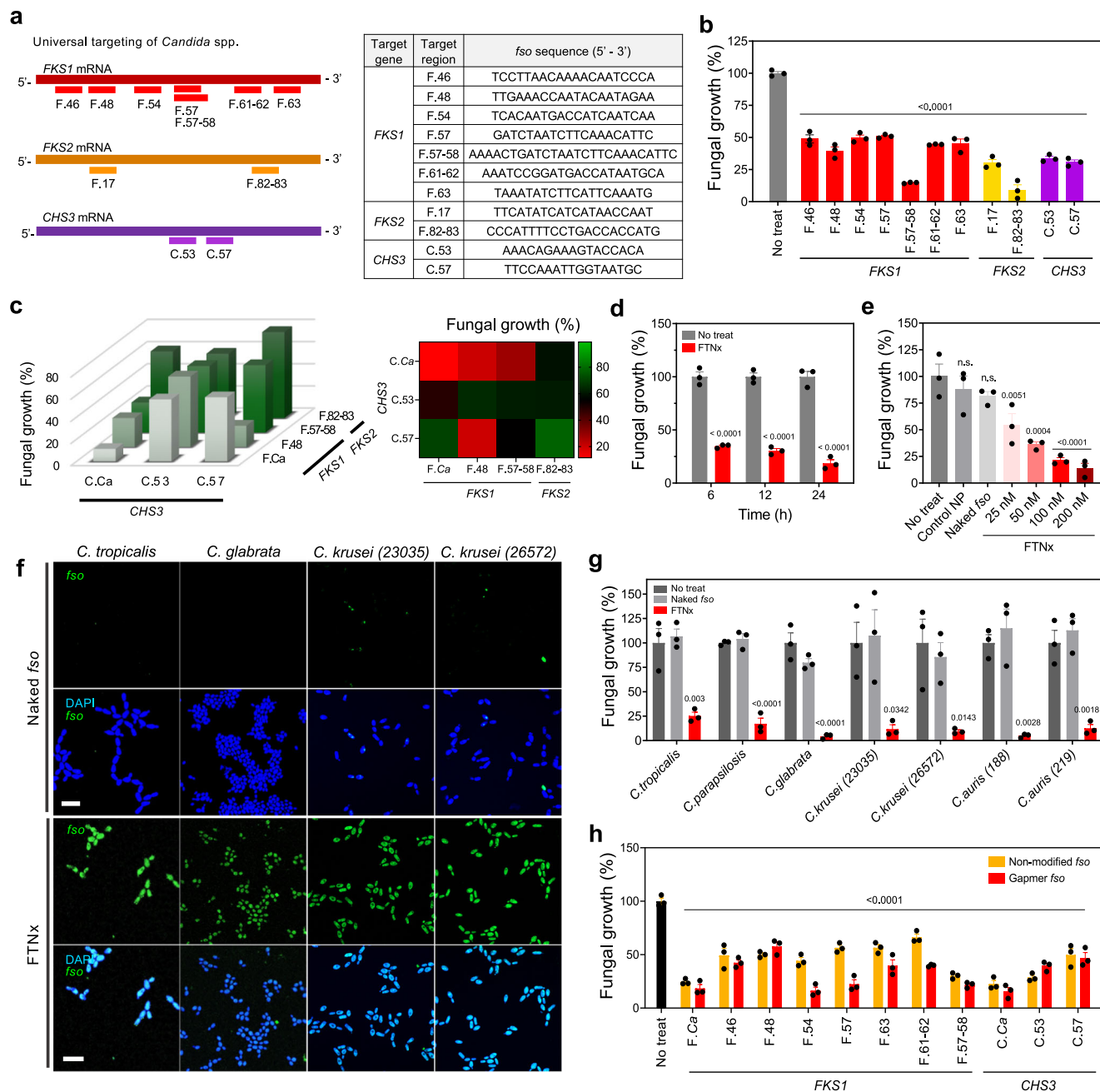


Fig. 4 | Development and validation of universal *fso* targeting multiple *Candida* species. **a** Design of candidate *fso*s targeting *FKS1*, *FKS2*, and *CHS3* to universally target various *Candida* spp. **b–e** Antifungal effects upon treatment of FTNx to *C. albicans* measured by the WST-8 assay. **b** Screening of the designed candidates by treating single *fso* via FTNx, and subsequently with micafungin (8 μg/ml). **c** Effect of dual *fso* delivery via FTNx, targeting either *FKS1* or *FKS2*, alongside *CHS3*. Heatmap depicts values of fungal growth. **d**, **e** Treatment of FTNx (F.48 + C.57) to *C. albicans* for various time periods (**d**); and at different concentrations of FTNx (**e**). **f**, **g** Treatment of FTNx (F.48 + C.57) to various *Candida* spp. and representative

confocal images of fungal uptake (**f**, 4 h; green, FAM-labeled *fso*; blue, DAPI; scale bar, 20 μm), and fungal growth (**g**, 24 h). **h** Antifungal effect upon treating FTNx with gapmer or unmodified single *fso* candidate to *C. albicans*, followed by treatment with micafungin (8 μg/ml). FTNx treated at 100 nM unless otherwise specified. **b–e**, **g**, **h** Data represent mean ± s.e.m. from experiments of biological replicates ($n = 3$). One-way ANOVA for comparison of treatment groups with 'No treat' control group. Representative data from three independent experiments. Source data are provided as a Source Data file.

(Mw 10 kDa, 765090) were purchased from Sigma-Aldrich. Chitosan (poly(D-glucosamine)) with high (190–310 kDa, CS_{high} , 448877), medium (50–190 kDa, CS_{med} , 448869), and low (5 kDa, CS_{low} , 523682-1) molecular weights were purchased from Sigma-Aldrich. Poly(allylamine) (PAA) (479144), poly(amidoamine) (PAM) (412392, 412414), spermine (SPN) (S3256), and protamine (PTN) (P3369) were obtained from Sigma-Aldrich. Unmodified and gapmer oligonucleotides used as *fso*, and primers were provided by Macrogen.

Preparation of FTNx and Nx(X) nanoconstructs

NHS-activated gold nanoparticles at a density of 1.1×10^{13} particles/ml in 1 mM NaCl were added with 1.0 μg/ml of BPEI (25 kDa) and incubated at room temperature (RT) with gentle stirring for 1 h. To remove the unreacted BPEI, the mixture was washed three times using centrifugal filters (Amicon® Ultracel®, MWCO 100 kDa, UFC810024) and resuspended in 10 mM NaCl. The resulting BPEI-functionalized gold nanoparticles (NP) were then combined with *fso* at a concentration of

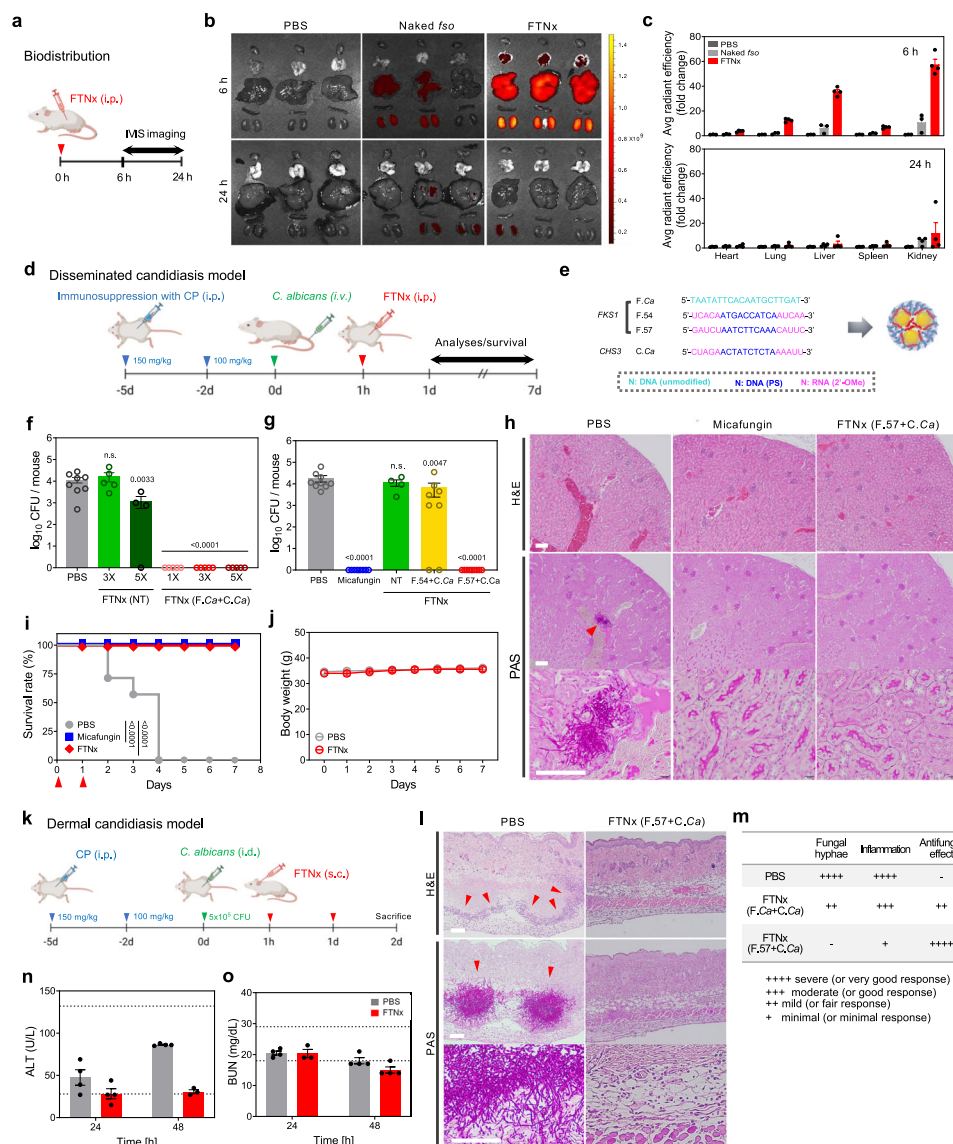


Fig. 5 | In vivo antifungal activity of FTNx. **a** Biodistribution studies in uninfected mice. **b** Fluorescence imaging of excised tissues after intraperitoneal injection of FTNx or naked *fso* (6–24 h). **c** Quantification of signals (Cy5.5-labeled *fso*) from (b). Data represent mean $\pm$ s.e.m. ($n = 4$ mice per group). **d** Antifungal efficacy in a disseminated candidiasis mouse model. **e** Sequences of *fso* for dual delivery. **f, g** Fungal cell burden (log₁₀ CFU) in infected mice after treatment. **f** Various doses of FTNx (F.Ca + C.Ca) (PBS, $n = 8$; FTNx (NT): 3X, $n = 5$; 5X, $n = 4$; FTNx (F.Ca + C.Ca): 1X, 3X, 5X, each $n = 5$ mice per group). **g** FTNx (3X, F.54 + C.Ca. and F.57 + C.Ca) compared with micafungin (10 mg/kg) and FTNx (NT) as controls (PBS, $n = 9$; micafungin, $n = 8$; FTNx (NT) 3X, $n = 4$; FTNx (F.54 + C.Ca), $n = 8$; FTNx (F.57 + C.Ca), $n = 9$ mice per group). **h** PAS and H&E staining of kidney tissues of treated mice. **i** Survival rate of mice in a lethal infection model treated twice with FTNx (3X, F.57 + C.Ca) at 24 h time interval ($n = 7$ mice). Arrowheads indicate injection time

points. **j** Body weight changes of uninfected mice after FTNx treatment ($n = 5$ mice per group). **k** Efficacy of FTNx in a dermal candidiasis model. **l** PAS and H&E staining of mouse skin tissues after treatment of FTNx (3X, F.57 + C.Ca) ($n = 4$ mice per group). **m** Examination scores for fungal pseudohyphae, inflammation, and antifungal effect from (l). **n, o** Biochemical analysis of blood serum from uninfected mice treated with FTNx (3X, F.57 + C.Ca). Dotted lines mark normal ranges of ALT and BUN levels ($n = 4$, except $n = 3$ for FTNx, 48 h (n) and FTNx, 24 h (o)). **b, h, l** Representative images shown from each group (scale bar, 100 μ m). **f, g, i, j, n, o** Data represent mean $\pm$ s.e.m. One-way ANOVA (f, g), two-way ANOVA (j), and log-rank (Mantel-Cox) test (i) for comparison with PBS control group. Source data are provided as a Source Data file. a, d, e, k created in BioRender. Chung, J. (2025) <https://BioRender.com/9q3cdpr>.

10–100 μ M and incubated at RT with gentle agitation for 30 min. After two washes using centrifugal filters, the resultant NP/*fso* complexes were resuspended in 10 mM NaCl, and then mixed with one of the cationic polymers or monomers (BPEI, LPEI, CS, PAA, PAM, SPN, or PTN) at a concentration of 2.0 μ g/ml and incubated at RT with gentle stirring for 1 h. Finally, the complexes were washed by centrifugal filtration and resuspended in 10 mM NaCl to obtain Nx(X).

For the preparation of the control formulations, PEI/*fso* and CS/*fso* complexes were prepared by mixing 50 nmol of *fso* with 1 μ g of BPEI and 2 μ g of CS, respectively, with gentle stirring for 2 h at RT. Silver

nanoparticle complexes (silver NC) was prepared as described for FTNx, using silver nanoparticles (10 nm, Cydiagnostics, S-10-20) as the core instead of gold nanoparticles, and CS as the cationic polymer. Cationic liposomes were produced using 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP, Avanti Polar Lipids, 890890 P) and 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE, Avanti Polar Lipids, 880233 C) at a molar ratio of 1:1⁶⁴. LNPs were prepared using the ionizable lipid D-Lin-MC3-DMA (MedKoo Biosciences, 555308) and helper lipids distearoylphosphatidylcholine (DSPC) (Avanti Polar Lipids, 850365 C), cholesterol (Avanti Polar Lipids, 700000 P), and

DMG-PEG2000 (Avanti Polar Lipids, 880151P) at a molar ratio of 50:10:38.5:1.5⁶⁵.

Physicochemical characterization of FTNx

DLS and zeta potential measurements were performed using ELSZ-2000 (Otsuka). For TEM, samples were stained with a 0.5% solution of ruthenium tetroxide (Sigma-Aldrich, 389000100), applied on a 300 mesh carbon grid (Electron Microscopy Sciences, Inc., CF-300 CU), and observed using Tecnai G2 Spirit TWIN (Thermo Fisher Scientific). For SEM, samples were dried overnight in a desiccator, subjected to platinum sputter coating, and observed using SU8230 (Hitachi) at a voltage of 10 kV.

Fungal strains and culture

Clinical strains of *C. albicans* AMC 27091, *C. glabrata* AMC 26571, *C. parapsilosis* AMC 27233, *C. krusei* AMC 23035 and AMC 26572, *C. auris* AMC 188 and AMC 219, and *C. tropicalis* AMC 27732 were provided by Asan Medical Center (AMC). For in vitro assays, the *Candida* strains were inoculated into Sabouraud Dextrose Broth (SDB, Difco™, BD, 238230) and incubated in suspension at 37 °C with gentle agitation (160–180 rpm) in a shaking incubator (Jeiotech) for 18 h. Fungal cells were then transferred to fresh media and incubated for another 2–3 h. Fungal growth was determined by measuring the optical densities at 600 nm (OD₆₀₀) using a NanoPhotometer® (P330, Implen GmbH).

For in vivo studies, *C. albicans* AMC 27091 was cultured on Chrom agar plates (BANDIO, MO2040) at 37 °C for 48 h, then a single colony was selected and inoculated into 5 mL of yeast extract peptone dextrose (YPD, Sigma-Aldrich, Y1500) medium. The cultures were incubated overnight at 37 °C with shaking at 180 rpm, followed by washing twice with sterile saline and resuspended to the appropriate cell concentration in sterile saline.

Cell culture

NIH-3T3 mouse embryonic fibroblast cells were obtained from the American Type Culture Collection (ATCC). Cell lines were authenticated by short tandem repeat profiling to confirm their identity. Cells were routinely tested and confirmed to be free of mycoplasma contamination. All experiments were conducted using cells at as low passage numbers as possible to minimize phenotypic drift. Cells were cultured in Dulbecco's Modified Eagle Medium (DMEM, Thermo Fisher Scientific, 11995073), including 25 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES, Gibco®, 15630080) and L-glutamine (Gibco®, Thermo Fisher Scientific, 25030081), supplemented with 10% heat-inactivated fetal bovine serum (FBS, Hyclone, SH30919.03) and 1% penicillin/streptomycin (P/S, Gibco®, Thermo Fisher Scientific, 15140122) in a CO₂ incubator at 37 °C (BB15, Thermo Fisher Scientific).

Uptake into fungal and mammalian cells

For examining fungal uptake, 10⁵ fungal cells were inoculated in microscopic chamber slides (SPL Life Science, 30108) and treated with either FTNx (100 nM) or control formulations including FAM-labeled *fso* (50 µM) for 2–4 h, followed by washing twice with PBS (pH 7.0). For mammalian cells, 2 × 10⁴ of NIH-3T3 cells were seeded in chamber slides and incubated with FTNx or control formulations for 4–6 h, followed by washing twice with PBS (pH 7.0). Then either fungal or mammalian cells were fixed in a 4% paraformaldehyde solution, permeabilized with 0.1% Triton X-100, and washed again twice with PBS. Then, fungal or mammalian cells were mounted on microscopic slides using VectaShield, including DAPI (Vector Laboratories, H-1200-10), and observed by confocal microscopy (LSM 980, Carl Zeiss). Z-stack imaging was performed to obtain cross-sectional views, to distinguish fluorescence signals in the cell wall and cytoplasm. Images were acquired and analyzed using Zen Blue software (Carl Zeiss).

Gene silencing effect of *fso*

Fungal cells of *C. albicans* (10⁵ CFU) were treated with FTNx (100 nM) loaded with *fso* for *FKSI* or *CHS3* gene at 37 °C with gentle stirring (180 rpm) for 2 h. Total RNA was extracted using TRIzol™ Reagent (Thermo Fisher Scientific, 15596018), and cDNA was synthesized using PrimeScript™ RT reagent kit (Takara Biotechnology, RR036A) with random primers. Quantitative polymerase chain reaction (qPCR) was then conducted with PowerUp™ SYBR™ Green (Thermo Fisher Scientific, A25780) and specific primers for each target gene (Supplementary Table 2) using QuantStudio 3 Real-Time PCR System (Thermo Fisher Scientific). The relative expression of *FKSI* and *CHS3* was determined by normalization to the expression levels of *ACT1*.

In vitro antifungal activity

The antifungal effect by FTNx was assessed using the Microbial Viability Assay Kit-WST (Dojindo Molecular Technologies, M439-10), according to the manufacturer's instructions. Briefly, cultured fungal cells were washed with PBS and treated at 10⁵ CFU/ml with FTNx (25–200 nM) or control formulations naked *fso*, or bare gold nanoparticles for 0–24 h, followed by exchange with fresh media with or without micafungin (1–8 µg/ml) and further incubation for 6–24 h. The WST reagent was then added, incubated at 37 °C for 1–2 h, and the absorbance was measured at 450 nm using Infinite® 200 PRO (Tecan, Ltd.). Fungal growth was determined by normalization with values for the control without treatment. For all in vitro experiments, equivalent amounts of *fso* in total were treated for FTNx and naked *fso* in each experimental set.

Live/dead assay

Cultured fungal cells were then diluted in fresh SDB medium to obtain an OD₆₀₀ of 0.4, and further incubated for 2.5 h. Then, 10⁵ fungal cells were treated with FTNx at 100 nM for 24 h at 37 °C and stained with the BacLight Bacterial Viability Kit (Invitrogen, L7007) according to the manufacturer's instructions. In brief, a 1:1 mixture of SYTO-9 and propidium iodide reagent was added to the fungal cells, incubated for 15 min at RT, and observed using a confocal microscope (LSM 980, Carl Zeiss).

Fluorescence microscopy of cell wall components

Fungal cells (3 × 10⁵ CFU) were treated with FTNx at 100 nM for 24 h, washed with PBS (pH 7.0), and fixed in 4% paraformaldehyde solution. Cells were then permeabilized using 0.1% Triton X-100, rinsed twice with PBS, and incubated in PBS containing 5% bovine serum albumin (BSA) for 30 min at RT, and washed three times with 0.5% BSA solution in PBS. For staining of β-1,3-glucan, Fc (human): Dectin-1 (mouse) (Adipogen, AG-40B-0138-C050, Lot. P27001708) was added to fungal cells at 1 µg/ml in PBS including 0.5% BSA, incubated on ice for 60 min, and washed three times with ice cold PBS including 0.5% BSA. Then, Alexa Fluor 488-conjugated anti-human IgG Fc (BioLegend, A22504, Lot. 3522103118) was added, incubated in ice for 45 min, and washed three times with PBS including 0.5% BSA. For staining mannan, Concanavalin A Texas Red Conjugate (ConA-TRed) (Life Technologies, C825) was added to fungal cells at 50 µg/mL in PBS, incubated in the dark at RT with gentle agitation for 45 min, and washed twice with PBS. For staining chitin, Calcofluor White (Sigma, 50 µg/mL, 18909) or WGA-AlexaFluor-488 conjugate (Invitrogen, 10 µg/mL, W11261) were added in PBS, and incubated for 10–30 min in the dark, followed by washing twice with PBS. The stained fungal cells were applied on microscopic chamber slides (SPL Life Science, 30108) and observed with a confocal microscope (LSM 980, Carl Zeiss).

Electron microscopy of fungal cells

For examining the morphology of fungal cells by TEM, fungal cells treated with FTNx were fixed in 0.1M sodium cacodylate buffer including 2% glutaraldehyde and 2% paraformaldehyde (pH 7.4) for

24 h at 4 °C, followed by washing with 0.1 M sodium cacodylate, and then post-fixed in 1% osmium tetroxide (Electron Microscopy Sciences, 50-275-72) for 1.5 h at 4 °C. After washing with 0.1 M sodium cacodylate, the samples were dehydrated in a graded ethanol series, propylene oxide, and then with 2:1, 1:1, and 1:2 mixture of propylene oxide and EMBed 812 resin (EMS). After embedding in pure EMBed 812 resin and polymerization at 60 °C for 24 h, ultrathin (70 nm) sections were prepared using an ultramicrotome (Leica, EM UC7) and applied on a 200-mesh formvar carbon film-coated copper grid (EMS, FCF200-Cu-50). Samples were finally stained with 2% uranyl acetate (7 min) and Reynold's lead citrate (3 min), observed with a transmission electron microscope (FEI, Tecnai G2 spirit TWIN) at 120 kV accelerating voltage. Images were acquired using a NanoSprint12 CMOS camera (Advanced Microscopy Techniques).

For SEM analysis, cultured cells were fixed in 2.5% glutaraldehyde in PBS (pH 7.0) at RT for 30 min, and then incubated for overnight at 4 °C, followed by washing twice with PBS. Samples were then treated with 1% osmium tetroxide (OsO₄) for 30 min, dehydrated in a graded ethanol series, and dried overnight in a desiccator. Samples were then analyzed using a scanning electron microscope (SU8230, Hitachi) at a voltage of 3–10 kV.

Animals

Male 7-week-old ICR mice were purchased from OrientBio. Animals were housed in a barrier environment with relative humidity of 50%. The average temperature was held around 20–22 °C. A 12/12 h light-dark cycle was maintained in the barrier environment. All animal studies were performed following authorized protocols and in accordance with the policies of the Institutional Animal Care and Use Committee of Asan Medical Center (#2022-12-260) and KAIST (#KA2023-076-v1).

In vivo biodistribution

Mice were anesthetized with isoflurane exposure and injected intraperitoneally with FTNx or naked *fso* (315 µg *fso*), which included 5'-Cy5.5 conjugated *fso*. Mice were then sacrificed at 6 h and 24 h, and tissues were excised and observed using the IVIS Lumina II imaging system.

Dermal candidiasis model

To induce immunosuppression, cyclophosphamide (CP) was administered intraperitoneally at a dose of 150 mg/kg, 4 days prior to infection, followed by a second dose of 100 mg/kg 1 day before infection⁵⁷. Three days before infection, mice were anesthetized, and the dorsal region was shaved for depilation. *C. albicans* (5×10^5 CFU) was then injected intradermally, and after one hour, mice were injected subcutaneously with PBS, micafungin (10 mg/kg; Sigma, SML2268), or FTNx (315 µg *fso* per mouse), and then another dose at 24 h post-infection. At 48 h post-infection, mice were euthanized, and the injection sites were examined by visualization. Skin tissues were excised, embedded in paraffin, and histological analyses were H&E PAS staining.

Disseminated candidiasis model

Immunosuppression was induced by intraperitoneal injection of CP at 150 mg/kg 5 days before infection, followed by a second dose of 100 mg/kg 2 days prior to infection. Mice were inoculated with *C. albicans* (5×10^4 CFU) via tail vein injection. One hour post-infection, mice were administered subcutaneously with micafungin (10 mg/kg), or intraperitoneally with PBS or FTNx (*fso* dosage; 1X, 105 µg; 3X, 315 µg; 5X, 525 µg per mouse). At 24 h post-infection, kidneys were harvested to assess the fungal burden. Histological analyses were performed by H&E and PAS staining.

To examine the survival rates in a lethal infection model, immunosuppression was induced by intraperitoneal administration

of CP at 150 mg/kg 4 days prior to infection, followed by a second dose of 100 mg/kg 1 day before infection. Mice were then inoculated with *C. albicans* (5×10^5 CFU) via tail vein injection. One hour post-infection, mice were administered subcutaneously with micafungin (10 mg/kg) or intraperitoneally with PBS or FTNx, followed by a second dose at 24 h post-infection. Survival of the mice was monitored for 7 days.

In vivo toxicity

The body weights and behavior of mice were monitored after treating with a single intraperitoneal injection of PBS or FTNx ($n = 2$) and observing for 7 days. To assess liver and kidney toxicity markers, mice were anesthetized with isoflurane and injected intraperitoneally with FTNx or the control ($n = 4$). At 24 h and 48 h, mice were sacrificed, and whole blood was collected by cardiac puncture, followed by measuring alanine aminotransferase (ALT) and blood urea nitrogen (BUN) levels using Catalyst One Chemistry Analyzer (IDEXX Laboratories).

Statistical analysis

For all in vitro experiments, at least three independent sets of experiments were performed in triplicate for each group, unless otherwise stated. Sample sizes were determined so that statistical analyses could be performed for comparison between groups. No statistical method was used to predetermine sample size. All statistical data were expressed as mean $\pm$ s.e.m., and analyses were carried out using GraphPad Prism v.9.0. Statistical significance was determined using one-way ANOVA, two-way ANOVA, or log-rank (Mantel-Cox) test to obtain the *p*-value. Exact *p*-values were reported in the figures or figure legends, and values less than 0.05 were considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data necessary to evaluate the conclusions of the study are provided in the article and/or the Supporting Information. Source data are provided with this paper and also available via Figshare (<https://doi.org/10.6084/m9.figshare.29085692.v1>). The sequence data of *fso* generated in this study are shown in the main figures and have been deposited in the Figshare database (<https://doi.org/10.6084/m9.figshare.29085623.v1>).

References

1. Brown, G. D. et al. The pathobiology of human fungal infections. *Nat. Rev. Microbiol.* **22**, 687–704 (2024).
2. Kainz, K., Bauer, M. A., Madeo, F. & Carmona-Gutierrez, D. Fungal infections in humans: the silent crisis. *Microb. Cell* **7**, 143–145 (2020).
3. Fisher, M. C. et al. Emerging fungal threats to animal, plant and ecosystem health. *Nature* **484**, 186–194 (2012).
4. de Oliveira Santos, G. C. et al. Candida infections and therapeutic strategies: Mechanisms of action for traditional and alternative agents. *Front. Microbiol.* **9**, 1351 (2018).
5. Lass-Flörl, C. et al. Invasive candidiasis. *Nat. Rev. Dis. Primers* **10**, 20 (2024).
6. Ahmadipour, S., Field, R. A. & Miller, G. J. Prospects for anti-Candida therapy through targeting the cell wall: A mini-review. *Cell Surf.* **7**, 100063 (2021).
7. Lee, Y., Puumala, E., Robbins, N. & Cowen, L. E. Antifungal drug resistance: Molecular mechanisms in *Candida albicans* and Beyond. *Chem. Rev.* **121**, 3390–3411 (2021).
8. Fisher, M. C. et al. Tackling the emerging threat of antifungal resistance to human health. *Nat. Rev. Microbiol.* **20**, 557–571 (2022).

9. Lockhart, S. R., Chowdhary, A. & Gold, J. A. W. The rapid emergence of antifungal-resistant human-pathogenic fungi. *Nat. Rev. Microbiol.* **21**, 818–832 (2023).
10. Berman, J. & Krysan, D. J. Drug resistance and tolerance in fungi. *Nat. Rev. Microbiol.* **18**, 319–331 (2020).
11. Izadi, A., Paknia, F., Roostaei, M., Mousavi, S. A. A. & Barani, M. Advancements in nanoparticle-based therapies for multidrug-resistant candidiasis infections: a comprehensive review. *Nanotechnology* **35**, 332001 (2024).
12. Nishikawa, J. L. et al. Inhibiting fungal multidrug resistance by disrupting an activator-Mediator interaction. *Nature* **530**, 485–489 (2016).
13. Netea, M. G., Joosten, L. A., van der Meer, J. W., Kullberg, B. J. & van de Veerdonk, F. L. Immune defence against *Candida* fungal infections. *Nat. Rev. Immunol.* **15**, 630–642 (2015).
14. He, J., Ye, Y., Zhang, D., Yao, K. & Zhou, M. Visualized gallium/lyticase-integrated antifungal strategy for fungal keratitis treatment. *Adv. Mater.* **34**, e2206437 (2022).
15. Oh, M. J. et al. Nanozyme-based robotics approach for targeting fungal infection. *Adv. Mater.* **36**, e2300320 (2024).
16. Lochenie, C. et al. Photosensitizer-amplified antimicrobial materials for broad-spectrum ablation of resistant pathogens in Ocular infections. *Adv. Mater.* **36**, e2404107 (2024).
17. Roberts, T. C., Langer, R. & Wood, M. J. A. Advances in oligonucleotide drug delivery. *Nat. Rev. Drug Discov.* **19**, 673–694 (2020).
18. Kumar, V., Huang, J., Dong, Y. & Hao, G. F. Targeting Fks1 proteins for novel antifungal drug discovery. *Trends Pharmacol. Sci.* **45**, 366–384 (2024).
19. Vaidya, A. et al. Expanding RNAi to Kidneys, Lungs, and Spleen via selective ORgan Targeting (SORT) siRNA lipid nanoparticles. *Adv. Mater.* **36**, e2313791 (2024).
20. Chen, Y., Zhu, X., Zhang, X., Liu, B. & Huang, L. Nanoparticles modified with tumor-targeting scFv deliver siRNA and miRNA for cancer therapy. *Mol. Ther.* **18**, 1650–1656 (2010).
21. Hou, X., Zaks, T., Langer, R. & Dong, Y. Lipid nanoparticles for mRNA delivery. *Nat. Rev. Mater.* **6**, 1078–1094 (2021).
22. Tenchov, R., Bird, R., Curtze, A. E. & Zhou, Q. Lipid Nanoparticles horizontal line From Liposomes to mRNA vaccine delivery, a landscape of research diversity and advancement. *ACS Nano* **15**, 16982–17015 (2021).
23. Angrish, N. & Khare, G. Antisense oligonucleotide based therapeutics and its applications against bacterial infections. *Med. Drug Discov.* **20**, 100166 (2023).
24. Matthew, D. D., Constantine, G. H. & Douglas, H. T. Uptake and antifungal activity of oligonucleotides in *Candida albicans*. *PNAS* **100**, 1530–1534 (2003).
25. Araujo, D. et al. Cationic lipid-based formulations for encapsulation and delivery of anti-EFG1 2' O-MethylRNA oligomer. *Med. Mycol.* **60**, myac030 (2022).
26. Araujo, D. et al. Polyamide micro-sized particulate polyplex carriers for the 2'-O-MethylRNA EFG1 antisense oligonucleotide. *ACS Appl. Bio. Mater.* **4**, 4607–4617 (2021).
27. Hu, X. et al. Structural and mechanistic insights into fungal beta-1,3-glucan synthase FKS1. *Nature* **616**, 190–198 (2023).
28. Ren, Z. et al. Structural basis for inhibition and regulation of a chitin synthase from *Candida albicans*. *Nat. Struct. Mol. Biol.* **29**, 653–664 (2022).
29. Yu, Q. et al. Inhibition of gold nanoparticles (AuNPs) on pathogenic biofilm formation and invasion to host cells. *Sci. Rep.* **6**, 26667 (2016).
30. M, N. et al. Studies on the antifungal activity of biotemplated gold nanoparticles over *Candida albicans*. *Mater. Res. Bull.* **119**, 110563 (2019).
31. McCarthy, J. R., Kelly, K. A., Sun, E. Y. & Weissleder, R. Targeted delivery of multifunctional magnetic nanoparticles. *Nanomedicine* **2**, 153–167 (2007).
32. Rosi, N. L. et al. Oligonucleotide-modified gold nanoparticles for intracellular gene regulation. *Science* **312**, 5776 (2006).
33. Beha, M. J., Ryu, J. S., Kim, Y. S. & Chung, H. J. Delivery of antisense oligonucleotides using multi-layer coated gold nanoparticles to methicillin-resistant *S. aureus* for combinatorial treatment. *Mater. Sci. Eng. C Mater. Biol. Appl.* **126**, 112167 (2021).
34. Fröhlich, E. The role of surface charge in cellular uptake and cytotoxicity of medical nanoparticles. *Int. J. Nanomed.* **7**, 5577 (2012).
35. Alburquenque, C. et al. Antifungal activity of low molecular weight chitosan against clinical isolates of *Candida* spp. *Med. Mycol.* **48**, 1018–1023 (2010).
36. Garcia, L. G. S. et al. Effect of the molecular weight of chitosan on its antifungal activity against *Candida* spp. in planktonic cells and biofilm. *Carbohydr. Polym.* **195**, 662–669 (2018).
37. Lee, S., Hao, L. T., Park, J., Oh, D. X. & Hwang, D. S. Nanochitin and nanochitosan: Chitin nanostructure engineering with multiscale properties for biomedical and environmental applications. *Adv. Mater.* **35**, e2203325 (2023).
38. Tayel, A. A. et al. Anticandidal action of fungal chitosan against *Candida albicans*. *Int. J. Biol. Macromol.* **47**, 454–457 (2010).
39. Pena, A., Sanchez, N. S. & Calahorra, M. Effects of chitosan on *Candida albicans*: conditions for its antifungal activity. *Biomed. Res. Int.* **2013**, 527549 (2013).
40. Whitfield, C. J. et al. Functional DNA-polymer conjugates. *Chem. Rev.* **121**, 11030–11084 (2021).
41. Hashem, A. H., Shehabeldine, A. M., Ali, O. M. & Salem, S. S. Synthesis of chitosan-based gold nanoparticles: Antimicrobial and wound-healing activities. *Polymers (Basel)* **14**, 2293 (2022).
42. Shih, P. Y., Liao, Y. T., Tseng, Y. K., Deng, F. S. & Lin, C. H. A potential antifungal effect of chitosan against *Candida albicans* is mediated via the inhibition of SAGA complex component expression and the subsequent alteration of cell surface integrity. *Front. Microbiol.* **10**, 602 (2019).
43. Foroozandeh, P. & Aziz, A. A. Insight into cellular uptake and intracellular trafficking of nanoparticles. *Nanoscale Res. Lett.* **13**, 339 (2018).
44. Behzadi, S. et al. Cellular uptake of nanoparticles: journey inside the cell. *Chem. Soc. Rev.* **46**, 4218–4244 (2017).
45. Salehi, Z. et al. Susceptibility pattern of caspofungin-coated gold nanoparticles against clinically important *Candida* species. *Adv. Pharm. Bull.* **11**, 693–699 (2021).
46. Lo, W. H., Deng, F. S., Chang, C. J. & Lin, C. H. Synergistic antifungal activity of chitosan with fluconazole against *Candida albicans*, *Candida tropicalis*, and fluconazole-resistant strains. *Molecules* **25**, 5114 (2020).
47. Walker, L. A. & Gow, N. A. Munro, ted chitin content reduces the susceptibility of *Candida* species to caspofungin. *Antimicrob. Agents Chemother.* **57**, 146–154 (2013).
48. Gagliardi, M. & Ashizawa, A. T. The challenges and strategies of antisense Oligonucleotide drug delivery. *Biomedicines* **9**, 433 (2021).
49. Chen, T. et al. Exposure of *Candida albicans* beta (1,3)-glucan is promoted by activation of the Cek1 pathway. *PLoS Genet.* **15**, e1007892 (2019).
50. Lee, Y., Robbins, N. & Cowen, L. E. Molecular mechanisms governing antifungal drug resistance. *NPJ Antimicrob. Resist.* **1**, 5 (2023).
51. Brand, A. Hyphal growth in human fungal pathogens and its role in virulence. *Int. J. Microbiol.* **2012**, 517529 (2012).
52. Reddy, G. K. K., Padmavathi, A. R. & Nancharaiiah, Y. V. Fungal infections: Pathogenesis, antifungals and alternate treatment approaches. *Curr. Res. Microb. Sci.* **3**, 100137 (2022).

53. Gow, N. A., Latge, J. P. & Munro, C. A. The fungal cell wall: structure, biosynthesis, and function. *Microbiol. Spectr.* **5.3**, 10–1128 (2017).
54. Smith, S. E. & Read, D. J. *Mycorrhizal Symbiosis*. (Academic Press, 2010).
55. Kilanowska, A. & Studzinska, S. In vivo and in vitro studies of anti-sense oligonucleotides - a review. *RSC Adv.* **10**, 34501–34516 (2020).
56. Duschmale, J. et al. In vitro and in vivo properties of therapeutic oligonucleotides containing non-chiral 3' and 5' thiophosphate linkages. *Nucleic Acids Res.* **48**, 63–74 (2020).
57. Conti, H. R., Huppler, A. R., Whibley, N. & Gaffen, S. L. Animal models for candidiasis. *Curr. Protoc. Immunol.* **105**, 1961 (2014).
58. Zuluaga, A. F. et al. Neutropenia induced in outbred mice by a simplified low-dose cyclophosphamide regimen: characterization and applicability to diverse experimental models of infectious diseases. *BMC Infect. Dis.* **6**, 55 (2006).
59. Backstrom, E. et al. Tissue pharmacokinetics of antisense oligonucleotides. *Mol. Ther. Nucleic Acids* **35**, 102133 (2024).
60. Geary, R. S., Norris, D., Yu, R. & Bennett, C. F. Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides. *Adv. Drug Deliv. Rev.* **87**, 46–51 (2015).
61. Zhang, S. et al. Prognostic value of liver and kidney function parameters and their correlation with the ratio of urine-to-plasma paraquat in patients with paraquat poisoning. *Basic Clin. Pharmacol. Toxicol.* **128**, 822–830 (2021).
62. Balcerrek, M. I., Stewart, A. G., Chapman, P. & Lazarus, S. Reducing the off-target endocrinologic adverse effects of azole antifungals—can it be done? *Int. J. Antimicrob. Agents* **59**, 106587 (2022).
63. Benjamin Lash, D., Jolliff, J., Munoz, A. & Heidari, A. Cross-reactivity between voriconazole, fluconazole and itraconazole. *J. Clin. Pharm. Ther.* **41**, 566–567 (2016).
64. Pereira, S. et al. Lipoplexes to deliver Oligonucleotides in Gram-Positive and Gram-Negative bacteria: Towards treatment of blood infections. *Pharmaceutics* **13**, 989 (2021).
65. Duskunovic, N., Im, S. H., Lee, J. & Chung, H. J. Effective mRNA delivery by condensation with cationic nanogels incorporated into liposomes. *Mol. Pharm.* **20**, 3088–3099 (2023).

Acknowledgements

This study was supported by grants from the Ministry of Health & Welfare (RS-2022-KH131278 to H.J.C.) and National Research Foundation (RS-2024-00355464 to H.J.C.), Republic of Korea. The authors acknowledge Yongsuk Hur (KAIST) for advice and technical assistance with electron microscopy and Daeyoup Lee (KAIST) for advice on and providing fungal strains.

Author contributions

J.Y.C., Y.-K.H., Y.P.C., and H.J.C. conceived and designed the experiments and wrote the manuscript. J.Y.C., Y.-K.H., and E.J. performed the experiments. J.Y.C., S.Y. and A.P. performed analyses on the data. R.W. provided valuable insights and critical review on the paper. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-025-60684-7>.

Correspondence and requests for materials should be addressed to Yong Pil Chong or Hyun Jung Chung.

Peer review information *Nature Communications* thanks Maurizio Del Poeta, Georgios Chamilos and Lingtong Meng for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025